PER SCHOLAS

SBA 928 - Enhancing Market Research with AI Prompt Engineering

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Introduction

Welcome to SBA 928 - Enhancing Market Research with AI Prompt Engineering! This assessment explores advanced AI prompt techniques to improve market research processes. You will engage in crafting prompts, fine-tuning language models, and addressing potential biases to optimize insights for business strategy and decision-making. Prepare to apply sophisticated AI practices to elevate market research effectiveness. This will help you gain insights regarding a data analyst role at a leading market research firm tasked with revolutionizing your company's approach to gathering and analyzing market insights.

Grading

Criteria	Exemplary (20 pts)	Proficient (10-19 pts)	Needs Improvement (0-5 pts)
Prompt Diversity and Relevance	Prompts are highly diverse, covering various aspects of market research comprehensively. They are well-tailored to elicit detailed and relevant insights.	Prompts cover several aspects of market research but may lack some diversity or relevance in certain areas.	Prompts are limited in scope, missing key aspects of market research or failing to generate relevant insights.
Fine-Tuning Effectiveness	Fine-tuning significantly improves the model's performance, with outputs showing enhanced relevance and accuracy in market research insights compared to the original model.	Fine-tuning results in some improvement in model performance, though differences from the original model may be moderate.	Fine-tuning shows minimal or no improvement in model performance, with outputs remaining similar to the original model.
Bias Detection and Mitigation	Comprehensive bias analysis is conducted, with effective strategies proposed for mitigating bias and ensuring fairness in market research insights.	Bias analysis is performed, with some strategies for mitigation proposed, though additional measures may be needed for full fairness.	Bias analysis is inadequate, with limited or no strategies proposed for addressing fairness in market research insights.
Quality of Analysis and Reporting	The report is thorough, well-structured, and provides clear insights into prompt generation, fine-tuning, and bias analysis. All deliverables are addressed with high quality and detail.	The report is complete and generally well-structured but may lack some detail or clarity in certain areas. Most deliverables are addressed adequately.	The report is incomplete, lacks detail, or is poorly structured. Several deliverables may be missing or inadequately addressed.

Objectives

- This assessment is designed to advance your skills in AI prompt engineering within the context of market research.
- It involves providing hands-on experience in creating prompts, fine-tuning language models, addressing potential bias, and analyzing strategies to ensure fairness.

The assessment will help develop a comprehensive understanding of how to leverage AI for actionable business insights, enhanced market research processes, and strategies for addressing potential biases to achieve accurate and equitable results.

Prerequisites

For this assessment, learners are expected to:

- Have a basic understanding of AI Prompt Engineering concepts.

Submission

Click the "Start Assignment" button on the Canvas Assignment page to upload your file. Make sure your submission includes clear explanations and well-documented code.

Please use this Per Scholas template to submit a report that should include:

- A curated dataset of diverse market research prompts (Requirement 1).
- Multiple prompt variations for different research scenarios (Requirement 2).
- Insights into the fine-tuning experiment and comparison with the original model (Requirements 3 and 4).
- Bias analysis and fairness strategies to ensure accurate and equitable market research insights (Requirements 5 and 6).

Requirements

Diverse AI Prompt Generation:

- **Sourcing a Dataset:**

  Use publicly available datasets related to market research from platforms such as Kaggle, Google Dataset Search, or UCI Machine Learning Repository. Alternatively, students can create a dataset by gathering publicly available market reports, surveys, or consumer data from reliable industry sources.

- **Create a Dataset:**

  Using the sourced Dataset, curate a collection of prompts that cover various aspects such as consumer behavior, market trends, and competitor analysis. The aim is to generate prompts that explore different dimensions of market research and gather comprehensive insights.

- **Prompt Variations:**

  Develop multiple variations of prompts from the dataset to elicit a range of responses. Discuss the approaches used to create these variations and how they can be applied to

different market research scenarios. This will highlight the ability of AI to provide diverse and nuanced insights.

Fine-Tuning Language Models:

- **Fine-Tuning Experiment:**

  Choose a language model and fine-tune it using the prompts developed in the previous requirement. Detail the fine-tuning process, including steps for data preparation, training, and adjustments made to enhance the model's performance in generating market research insights.

- **Comparison with Original Model:**

  Compare the outputs of the fine-tuned model with those of the original model. Analyze the improvements in the relevance and accuracy of market research insights and assess the effectiveness of the fine-tuning process.

Bias Analysis and Fairness Strategies:

- **Bias Assessment:**

  Evaluate the prompts for any potential biases that might affect the quality and fairness of market research insights. Discuss methods for detecting and mitigating biases to ensure that the AI-generated data reflects diverse and equitable perspectives.

- **Strategies for Fairness:**

  Propose strategies and techniques to ensure fairness in prompt generation for market research. Consider the various aspects of market research such as demographic diversity, regional differences, and industry-specific contexts to ensure comprehensive and unbiased insights.

Time/Duration

This assessment will be conducted over one day.